

FINAL PROJECT • STUDENT GUIDE

PROJECT GUIDANCE FOR STUDENTS

1501418 Selected Topics in Data Science and AI

Assessment: Final Project

Project Weight: 50% of the total course grade

Project Type: Group project

Recommended Group Size: 4-8 students

Problem → Data → Model → Decision → Action

1. Project Overview

The final project requires students to apply concepts from data science, artificial intelligence, and decision modeling to a real-world problem.

Each group must:

1. Select and justify a real-world problem using an available dataset.
2. Examine the dataset and design an appropriate data model.
3. Propose a decision model that uses the data to support prediction, classification, recommendation, optimization, planning, or another practical decision.

The project should demonstrate the complete relationship between:

Problem → Dataset → Data Model → Analytical/AI Model → Decision → Action

The proposed solution may use one or more approaches covered in the course:

- Statistical analysis
- Machine learning
- Optimization
- Generative AI
- Large Language Models
- Retrieval-Augmented Generation
- Hybrid decision models

2. Project Learning Outcomes

After completing the project, students should be able to:

1. Identify a practical problem that can be addressed using data.
2. Explain why the problem is important to relevant stakeholders.
3. Select and evaluate an appropriate dataset.
4. Design a data model to organize and manage the selected data.
5. Select an appropriate statistical, AI, or optimization method.
6. Translate model outputs into practical decisions.
7. Define suitable evaluation measures and success criteria.
8. Communicate the project clearly through a report, presentation, and demonstration.

PART 1: PROBLEM IDENTIFICATION AND JUSTIFICATION

Weight: 10 Marks

3.1 Select a Real-World Case

Each group must select a case from a real organization, industry, community, public service, or research area.

Possible application areas include:

- Agriculture and smart farming
- Business and marketing
- Finance and investment
- Healthcare
- Education
- Logistics and transportation
- Tourism
- Public safety
- Environmental monitoring
- Disaster-risk management
- Customer service
- Human-resource management

The case must have an identifiable dataset that can be accessed legally and ethically.

3.2 Define the Problem

Clearly explain:

- What is happening?
- Who experiences the problem?
- What decision currently needs to be made?
- What limitations exist in the current process?
- How could data or AI improve the decision?

Use the following statement to define the decision problem:

The proposed model will help _____ to decide, predict, recommend, classify, or optimize _____ by using _____ data.

Example

The proposed model will help durian growers decide when irrigation is required by using soil-moisture, temperature, humidity, and weather data.

3.3 Justify the Problem

The justification should explain why the selected problem is worth solving.

Support the explanation with evidence such as:

- Statistics
- Research findings
- Industry or government reports
- Organizational information
- Stakeholder requirements
- Economic impacts
- Social or environmental impacts

Students should not rely only on personal opinions.

3.4 Identify Stakeholders

Identify the people or organizations involved in the problem.

Stakeholder	Role in the problem	Information or decision required
Example: Farm owner	Decision maker	Whether and when to irrigate
Example: Farm worker	System user	Required irrigation action
Example: Agricultural expert	Knowledge provider	Irrigation rules and thresholds

3.5 Define the Project Objectives

Provide two to four measurable objectives.

Example:

1. To examine the relationship between environmental conditions and irrigation requirements.
2. To develop a model for classifying irrigation conditions.
3. To propose decision rules for recommending irrigation actions.

3.6 Define the Project Questions

Possible questions include:

- Which variables are most relevant to the selected problem?
- What patterns can be discovered from the dataset?
- Which model is appropriate for supporting the decision?
- How can the model output be converted into a practical action?
- How will the model be evaluated?

Part 1 Assessment

Assessment criterion	Marks
Clarity of the case and problem context	2
Identification of stakeholders and the decision problem	2
Evidence-based problem justification	3
Clarity of objectives, questions, and expected outcomes	3
Part 1 Total	10

PART 2: DATASET AND DATA MODEL

Weight: 15 Marks

4.1 Select the Dataset

Students may use:

- Public or open datasets
- Government datasets
- Research datasets
- Organizational datasets with permission
- Survey data
- Sensor and Internet of Things data
- Transaction data
- Images, audio, video, or text
- Geographic or spatial data
- Time-series data
- Multiple related datasets

4.2 Provide Dataset Information

Complete the following table.

Dataset item	Project information
Dataset name	
Dataset provider or owner	
Dataset source or URL	
Data-collection period	
Number of records	
Number of attributes	
Data format	
Target variable, if applicable	
License or usage conditions	
Date accessed	

4.3 Explain Dataset Relevance

Explain:

- Why the dataset is suitable for the problem
- How the variables relate to the decision
- Whether the data are sufficiently complete
- Whether the data represent the intended population or situation
- What limitations or biases may exist

4.4 Create a Data Dictionary

Describe the important attributes in the dataset.

Variable	Data type	Description	Example	Role
Customer_ID	Integer	Unique customer identifier	10001	Identifier
Monthly_Usage	Numeric	Monthly service usage	48.5	Input feature
Contract_Type	Categorical	Type of service contract	Annual	Input feature
Churn	Binary	Whether the customer left	1	Target

The role of each variable may be:

- Identifier
- Input feature
- Target variable
- Decision variable
- Constraint
- Metadata
- Variable excluded from the model

4.5 Assess Data Quality

The group must examine:

- Missing values
- Duplicate records
- Incorrect or inconsistent values

- Outliers
- Imbalanced classes
- Data bias
- Sensitive or personal information
- Data types and formats
- Required cleaning and transformation

Explain how each identified problem will be handled.

4.6 Conduct Exploratory Data Analysis

Provide at least three meaningful analyses or visualizations.

Examples include:

- Distribution of important variables
- Comparison between categories
- Correlation between variables
- Time-series trends
- Geographic patterns
- Target-class distribution
- Image or text characteristics
- Relationships between input variables and the target

Each visualization must include:

1. A descriptive title
2. Clearly labeled axes or categories
3. A short interpretation
4. An explanation of its relevance to the decision problem

4.7 Design the Data Model

Design a model showing how the data will be organized and used.

Depending on the project, the data model may be:

- Entity-Relationship Diagram
- Relational database schema
- Star schema

- Feature table
- Data pipeline
- Knowledge graph
- Document and metadata structure
- Vector database for a RAG system

The proposed data model should identify:

- Entities, tables, or collections
- Attributes
- Primary and foreign keys, where applicable
- Relationships between data elements
- Input features
- Target variables
- Data transformations
- Data flow from source to model

Part 2 Assessment

Assessment criterion	Marks
Appropriateness and reliability of the dataset	3
Dataset description and data dictionary	3
Data-quality assessment and preprocessing plan	3
Exploratory data analysis and interpretation	3
Correctness and clarity of the data model	3
Part 2 Total	15

PART 3: PROPOSED DECISION MODEL

Weight: 25 Marks

5.1 Select the Model Type

Select a model that is appropriate for the decision problem.

Model category	Possible applications
Statistical model	Risk analysis, factor analysis, hypothesis testing
Classification model	Fraud, disease, risk, or customer-churn classification
Regression model	Price, demand, yield, or sales prediction
Clustering model	Customer, farm, product, or community segmentation
Recommendation model	Product, content, treatment, or activity recommendation
Optimization model	Scheduling, transportation, production, or resource allocation
Time-series model	Demand, weather, price, or environmental forecasting
Generative AI/LLM	Report generation, summarization, and question answering
RAG model	Evidence-based answers using external documents
Hybrid model	Combination of rules, ML, optimization, and generative AI

Students must explain why the selected model is more suitable than possible alternatives.

5.2 Define the Decision Model

Complete the following structure.

Component	Description
Model users	Who will use the model?
Input	What data will enter the model?
Processing	What algorithm, method, or rules will be used?
Output	What result will the model produce?
Decision	What decision will be supported?
Action	What will the user do after receiving the result?
Feedback	How will actual results improve the model?

5.3 Develop the Model Workflow

The proposed workflow should include:

1. Data collection
2. Data validation
3. Data cleaning and preprocessing
4. Feature selection or feature engineering
5. Model training or formulation
6. Model validation and testing
7. Result interpretation
8. Decision recommendation
9. User action
10. Feedback and model improvement

Students should present the workflow using a clear diagram.

5.4 Explain the Technical Design

For Statistical or Machine-Learning Models

Specify:

- Input features
- Target variable
- Selected algorithm
- Baseline method
- Training, validation, and testing strategy
- Parameter or hyperparameter selection
- Expected model output

For Optimization Models

Specify:

- Decision variables
- Objective function
- Constraints
- Input parameters
- Assumptions

- Expected optimal solution

For Generative AI or RAG Models

Specify:

- Source documents or knowledge base
- Document preparation and chunking
- Embedding method
- Vector database
- Retrieval process
- Prompt design
- Response-generation process
- Citation or grounding method
- Hallucination-control measures

For Hybrid Models

Explain how the selected methods are connected and how information moves between the models.

5.5 Define Model Evaluation

Use measures that match the selected task.

Model task	Possible evaluation measures
Classification	Accuracy, Precision, Recall, F1-score, ROC-AUC
Regression	MAE, MSE, RMSE, R ²
Clustering	Silhouette score and expert interpretation
Recommendation	Precision@K, Recall@K, user satisfaction
Optimization	Objective value, cost reduction, time saved, resource utilization
Forecasting	MAE, RMSE, MAPE
LLM/RAG	Correctness, relevance, faithfulness, groundedness, citation quality
Decision-support system	Usefulness, usability, acceptance, decision improvement

Students must define a measurable success criterion.

Example:

The model will be considered successful if it achieves an F1-score of at least 0.80 and the recommendations are judged useful by at least 80% of test users.

5.6 Translate the Output into a Decision

The model output must lead to a practical decision or action.

Example:

Model output	Decision level	Recommended action
Risk probability below 0.40	Low	Continue normal monitoring
Risk probability 0.40-0.69	Medium	Increase monitoring and notify the responsible person
Risk probability 0.70 or higher	High	Take immediate preventive action

Students must explain:

- Who receives the result
- How the result will be displayed
- What action should be taken
- Whether human approval is required
- How the action will be recorded

5.7 Develop a Proof of Concept

Each group must provide at least one of the following:

- Working model prototype
- Jupyter Notebook
- Model simulation
- Optimization spreadsheet
- Interactive dashboard
- RAG demonstration
- User-interface prototype
- Model pseudocode with sample outputs

The proof of concept does not need to be a complete production system, but it must demonstrate how the proposed model would work.

5.8 Explain Benefits, Limitations, and Risks

Discuss the expected benefits, such as:

- Improved decision accuracy

- Reduced cost
- Faster decision-making
- Better resource allocation
- Reduced risk
- Improved service quality
- Social or environmental benefits

Also discuss:

- Dataset limitations
- Model assumptions
- Bias and fairness
- Privacy and confidentiality
- Model explainability
- Incorrect predictions
- Generative AI hallucinations
- Security risks
- Need for human oversight

Part 3 Assessment

Assessment criterion	Marks
Appropriateness and justification of the selected model	4
Completeness of the model architecture and workflow	4
Technical correctness of the model formulation	4
Evaluation plan, baseline, and success criteria	4
Translation of model results into decisions and actions	3
Quality of the proof of concept or demonstration	3
Benefits, limitations, ethical considerations, and risks	3
Part 3 Total	25

6. Project Deliverables

6.1 Project Report

- Recommended length: 10-15 pages
- File format: PDF
- References: APA 7th edition
- Appendices are not included in the page limit.

6.2 Dataset and Supporting Files (Appendix in report)

Submit:

- Dataset or accessible dataset link
- Data dictionary
- Data-cleaning or preprocessing files
- Source code or notebook
- Data model diagram
- Decision-model diagram
- Demonstration files or application link

6.3 Presentation Slides

- Approximately 12-15 slides
- Presentation time: 15 minutes
- Question-and-answer session: 5 minutes
- Every group member must participate.

7. Recommended Report Structure

1. Cover Page
2. Executive Summary
3. Problem Background
4. Problem Justification
5. Stakeholders and Decision Requirements
6. Project Objectives and Questions
7. Dataset Description
8. Data Dictionary
9. Data Quality and Preprocessing

10. Exploratory Data Analysis
11. Data Model
12. Proposed Decision Model
13. Model Architecture and Workflow
14. Evaluation Plan and Success Criteria
15. Decision Rules and Recommended Actions
16. Proof of Concept (Data visualization / Engine / Model simulation .. etc)
17. Expected Benefits
18. Limitations, Ethics, and Risks
19. Conclusion
20. References
21. Appendices

8. Recommended Presentation Structure

Slide	Recommended content
1	Project title and group members
2	Problem background
3	Problem justification and evidence
4	Stakeholders and decision problem
5	Objectives and project questions
6	Dataset overview
7	Data quality and preprocessing
8	Exploratory data analysis
9	Data model
10	Proposed decision model / Visualization
11	Model workflow or architecture
12	Evaluation measures and success criteria
13	Decision rules and recommended actions
14	Proof-of-concept demonstration
15	Benefits, limitations, and conclusion

9. Overall Project Scoring

Project component	Marks	Contribution to course grade
Part 1: Problem Identification and Justification	10	10%
Part 2: Dataset and Data Model	15	15%
Part 3: Proposed Decision Model	25	25%
Total Project Score	50	50%

10. Performance Levels

Level	Description
Excellent	The project is clear, evidence-based, technically correct, feasible, and strongly connected to a real decision.
Good	The project is mostly complete and correct, with minor weaknesses in evidence, analysis, or model design.
Satisfactory	The basic requirements are met, but the analysis, justification, or decision model lacks depth.
Needs Improvement	The work is incomplete, unclear, technically inappropriate, or weakly connected to a practical decision.

11. Academic Integrity and Responsible AI Use

Students must follow these requirements:

1. Properly cite all datasets, reports, research articles, models, software, and external sources.
2. Do not present simulated or fabricated data as real data.
3. Obtain permission before using private organizational data.
4. Remove or anonymize sensitive and personally identifiable information.
5. Clearly acknowledge the use of generative AI tools.
6. Verify all AI-generated content, code, references, and analytical results.
7. Do not submit AI-generated work without human review and meaningful student contribution.
8. Each student must be able to explain the dataset, data model, analytical method, and decision model.
9. Undeclared copying, plagiarism, or fabrication may result in a score of zero.

12. Final Submission Checklist

Before submitting the project (**1.Report 2. Presentation slides**), confirm that the group has included:

- ☐ A clearly defined real-world problem
- ☐ Evidence supporting the importance of the problem
- ☐ Identified stakeholders and decision requirements
- ☐ Clear objectives and project questions
- ☐ A reliable and properly cited dataset
- ☐ A complete data dictionary
- ☐ Data-quality and preprocessing analysis
- ☐ At least three meaningful data visualizations
- ☐ A clear data model
- ☐ A justified decision-model approach
- ☐ A model architecture or workflow diagram
- ☐ Appropriate evaluation measures
- ☐ An individual contribution statement for every member
- ☐ Measurable success criteria
- ☐ Decision rules connecting model outputs to actions
- ☐ A proof of concept or demonstration
- ☐ Discussion of benefits, limitations, ethics, and risks
- ☐ Complete references in APA 7th edition